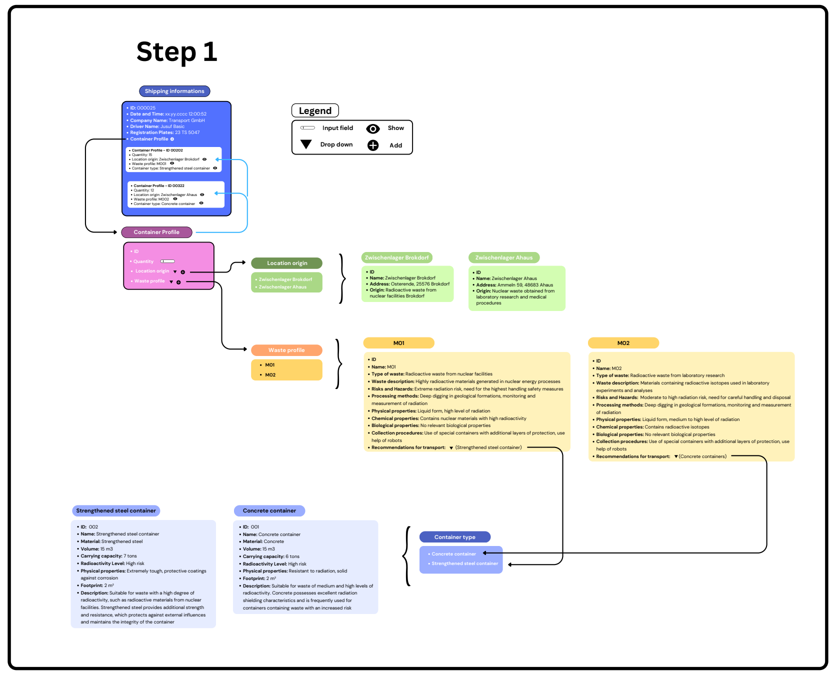
Step 1



Shipping informations

- **ID:** 000025
- **Date and Time:** xx.yy.cccc 12:00:52
- **Company Name:** Transport GmbH
- **Driver Name:** Jusuf Basic
- **Registration Plates:** 23 TS 5047
- **Container Profile** ⊕
  - **Container Profile - ID 00202**
  - Quantity: 15
  - Location origin: Zwischenlager Brokdorf 👁
  - Waste profile: M001 👁
  - Container type: Strengthened steel container 👁
  - **Container Profile - ID 00322**
  - Quantity: 12
  - Location origin: Zwischenlager Ahaus 👁
  - Waste profile: M002 👁
  - Container type: Concrete container 👁

Legend

▭	Input field	👁	Show
▼	Drop down	⊕	Add

Container Profile

- ID
- Quantity ▭
- Location origin ▼ ⊕
- Waste profile ▼ ⊕

Location origin

- Zwischenlager Brokdorf
- Zwischenlager Ahaus

Zwischenlager Brokdorf

- ID
- **Name:** Zwischenlager Brokdorf
- **Address:** Osterende, 25576 Brokdorf
- **Origin:** Radioactive waste from nuclear facilities Brokdorf

Zwischenlager Ahaus

- ID
- **Name:** Zwischenlager Ahaus
- **Address:** Ammeln 59, 48683 Ahaus
- **Origin:** Nuclear waste obtained from laboratory research and medical procedures

Waste profile

- M01
- M02

M01

- ID
- **Name:** M01
- **Type of waste:** Radioactive waste from nuclear facilities
- **Waste description:** Highly radioactive materials generated in nuclear energy processes
- **Risks and Hazards:** Extreme radiation risk, need for the highest handling safety measures
- **Processing methods:** Deep digging in geological formations, monitoring and measurement of radiation
- **Physical properties:** Liquid form, high level of radiation
- **Chemical properties:** Contains nuclear materials with high radioactivity
- **Biological properties:** No relevant biological properties
- **Collection procedures:** Use of special containers with additional layers of protection, use help of robots
- **Recommendations for transport:** ▼ (Strengthened steel container)

M02

- ID
- **Name:** M02
- **Type of waste:** Radioactive waste from laboratory research
- **Waste description:** Materials containing radioactive isotopes used in laboratory experiments and analyses
- **Risks and Hazards:** Moderate to high radiation risk, need for careful handling and disposal
- **Processing methods:** Deep digging in geological formations, monitoring and measurement of radiation
- **Physical properties:** Liquid form, medium to high level of radiation
- **Chemical properties:** Contains radioactive isotopes
- **Biological properties:** No relevant biological properties
- **Collection procedures:** Use of special containers with additional layers of protection, use help of robots
- **Recommendations for transport:** ▼(Concrete containers)

Strengthened steel container

- **ID:** 002
- **Name:** Strengthened steel container
- **Material:** Strengthened steel
- **Volume:** 15 m3
- **Carrying capacity:** 7 tons
- **Radioactivity Level:** High risk
- **Physical properties:** Extremely tough, protective coatings against corrosion
- **Footprint:** 2 m²
- **Description:** Suitable for waste with a high degree of radioactivity, such as radioactive materials from nuclear facilities. Strengthened steel provides additional strength and resistance, which protects against external influences and maintains the integrity of the container

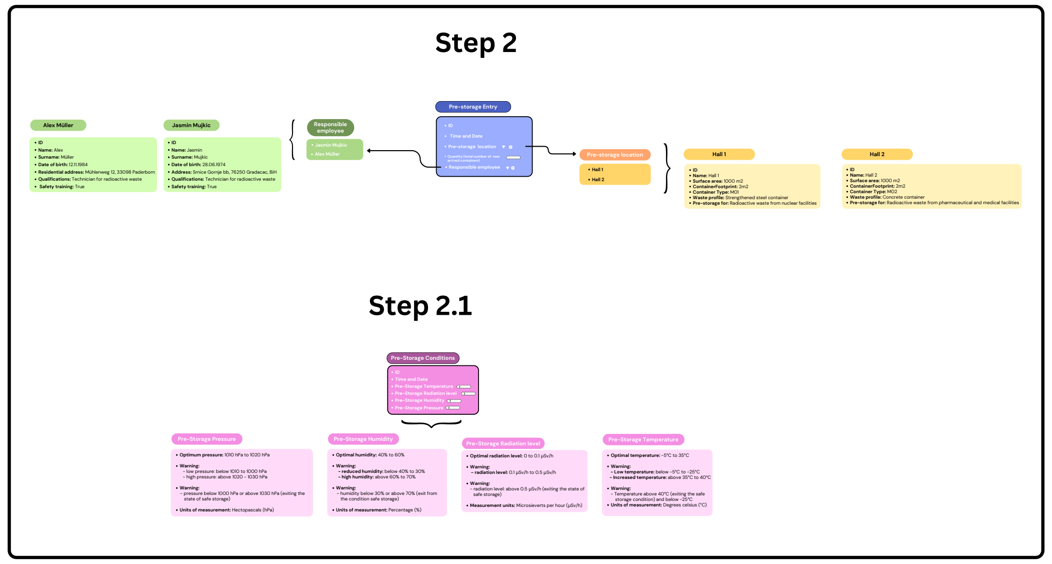
Concrete container

- **ID:** 001
- **Name:** Concrete container
- **Material:** Concrete
- **Volume:** 15 m3
- **Carrying capacity:** 6 tons
- **Radioactivity Level:** High risk
- **Physical properties:** Resistant to radiation, solid
- **Footprint:** 2 m²
- **Description:** Suitable for waste of medium and high levels of radioactivity. Concrete possesses excellent radiation shielding characteristics and is frequently used for containers containing waste with an increased risk

Container type

- Concrete container
- Strengthened steel container

Step 2



Alex Müller

- ID
- **Name:** Alex
- **Surname:** Müller
- **Date of birth:** 12.11.1984
- **Residential address:** Mühlenweg 12, 33098 Paderborn
- **Qualifications:** Technician for radioactive waste
- **Safety training:** True

Jasmin Mujkic

- ID
- **Name:** Jasmin
- **Surname:** Mujkic
- **Date of birth:** 28.06.1974
- **Address:** Srnice Gornje bb, 76250 Gradacac, BiH
- **Qualifications:** Technician for radioactive waste
- **Safety training:** True

Responsible employee

- Jasmin Mujkic
- Alex Müller

Pre-storage Entry

- ID
- Time and Date
- Pre-storage location ▼ ⊕
- Quantity (total number of new arrived containers) ▭
- Responsible employee ▼ ⊕

Pre-storage location

- Hall 1
- Hall 2

Hall 1

- ID
- **Name:** Hall 1
- **Surface area:** 1000 m2
- **ContainerFootprint:** 2m2
- **Container Type:** M01
- **Waste profile:** Strengthened steel container
- **Pre-storage for:** Radioactive waste from nuclear facilities

Hall 2

- ID
- **Name:** Hall 2
- **Surface area:** 1000 m2
- **ContainerFootprint:** 2m2
- **Container Type:** M02
- **Waste profile:** Concrete container
- **Pre-storage for:** Radioactive waste from pharmaceutical and medical facilities

Step 2.1

Pre-Storage Conditions

- ID
- Time and Date
- Pre-Storage Temperature ▭
- Pre-Storage Radiation level ▭
- Pre-Storage Humidity ▭
- Pre-Storage Pressure ▭

Pre-Storage Pressure

- **Optimum pressure:** 1010 hPa to 1020 hPa
- **Warning:**
  - low pressure: below 1010 to 1000 hPa
  - high pressure: above 1020 - 1030 hPa
- **Warning:**
  - pressure below 1000 hPa or above 1030 hPa (exiting the state of safe storage)
- **Units of measurement:** Hectopascals (hPa)

Pre-Storage Humidity

- **Optimal humidity:** 40% to 60%
- **Warning:**
  - **reduced humidity:** below 40% to 30%
  - **high humidity:** above 60% to 70%
- **Warning:**
  - humidity below 30% or above 70% (exit from the condition safe storage)
- **Units of measurement:** Percentage (%)

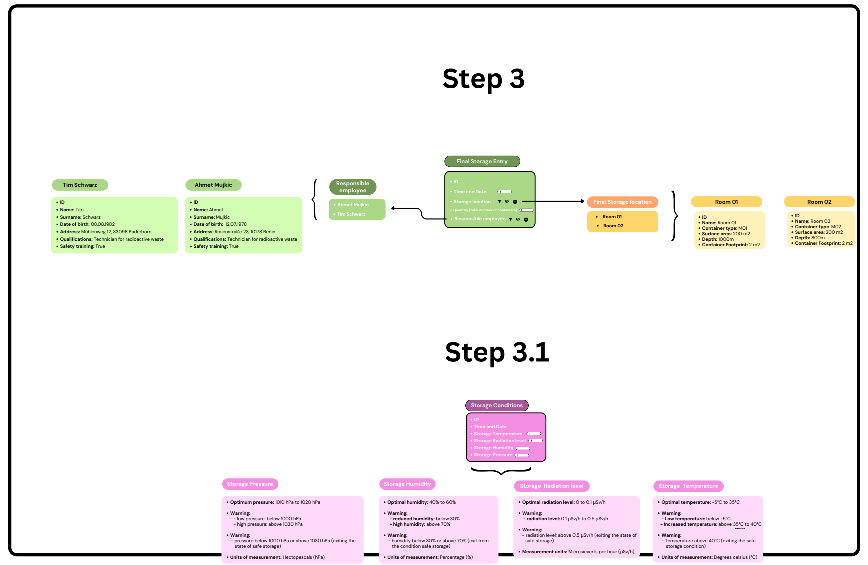
Pre-Storage Radiation level

- **Optimal radiation level:** 0 to 0.1 μSv/h
- **Warning:**
  - **radiation level:** 0.1 μSv/h to 0.5 μSv/h
- **Warning:**
  - radiation level: above 0.5 μSv/h (exiting the state of safe storage)
- **Measurement units:** Microsieverts per hour (μSv/h)

Pre-Storage Temperature

- **Optimal temperature:** -5°C to 35°C
- **Warning:**
  - **Low temperature:** below -5°C to -25°C
  - **Increased temperature:** above 35°C to 40°C
- **Warning:**
  - Temperature above 40°C (exiting the safe storage condition) and below -25°C
- **Units of measurement:** Degrees celsius (°C)

Step 3



Tim Schwarz

- ID
- **Name:** Tim
- **Surname:** Schwarz
- **Date of birth:** 08.08.1982
- **Address:** Mühlenweg 12, 33098 Paderborn
- **Qualifications:** Technician for radioactive waste
- **Safety training:** True

Ahmet Mujkic

- ID
- **Name:** Ahmet
- **Surname:** Mujkic
- **Date of birth:** 12.07.1978
- **Address:** Rosenstraße 23, 10178 Berlin
- **Qualifications:** Technician for radioactive waste
- **Safety training:** True

Responsible employee

- Ahmet Mujkic
- Tim Schwarz

Final Storage Entry

- ID
- Time and Date ▭
- Storage location ▼ 👁 ⊕
- Quantity (total number of containers) ▭
- Responsible employee ▼ 👁 ⊕

Final Storage location

- Room 01
- Room 02

Room 01

- ID
- **Name:** Room 01
- **Container type:** M01
- **Surface area:** 200 m2
- **Depth:** 1000m
- **Container Footprint:** 2 m2

Room 02

- ID
- **Name:** Room 02
- **Container type:** M02
- **Surface area:** 200 m2
- **Depth:** 800m
- **Container Footprint:** 2 m2

Step 3.1

Storage Conditions

- ID
- Time and Date
- Storage Temperature ▭
- Storage Radiation level ▭
- Storage Humidity ▭
- Storage Pressure ▭

Storage Pressure

- **Optimum pressure:** 1010 hPa to 1020 hPa
- **Warning:**
  - low pressure: below 1000 hPa
  - high pressure: above 1030 hPa
- **Warning:**
  - pressure below 1000 hPa or above 1030 hPa (exiting the state of safe storage)
- **Units of measurement:** Hectopascals (hPa)

Storage Humidity

- **Optimal humidity:** 40% to 60%
- **Warning:**
  - **reduced humidity:** below 30%
  - **high humidity:** above 70%
- **Warning:**
  - humidity below 30% or above 70% (exit from the condition safe storage)
- **Units of measurement:** Percentage (%)

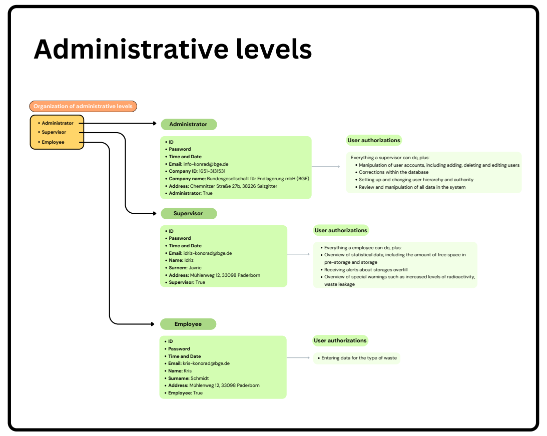
Storage Radiation level

- **Optimal radiation level:** 0 to 0.1 μSv/h
- **Warning:**
  - **radiation level:** 0.1 μSv/h to 0.5 μSv/h
- **Warning:**
  - radiation level: above 0.5 μSv/h (exiting the state of safe storage)
- **Measurement units:** Microsieverts per hour (μSv/h)

Storage Temperature

- **Optimal temperature:** -5°C to 35°C
- **Warning:**
  - **Low temperature:** below -5°C
  - **Increased temperature:** above 35°C to 40°C
- **Warning:**
  - Temperature above 40°C (exiting the safe storage condition)
- **Units of measurement:** Degrees celsius (°C)

Administrative levels



Organization of administrative levels

- Administrator
- Supervisor
- Employee

Administrator

- ID
- Password
- Time and Date
- **Email:** info-konrad@bge.de
- **Company ID:** 1651-3131531
- **Company name:** Bundesgesellschaft für Endlagerung mbH (BGE)
- **Address:** Chemnitzer Straße 27b, 38226 Salzgitter
- **Administrator:** True

User authorizations

Everything a supervisor can do, plus:

- Manipulation of user accounts, including adding, deleting and editing users
- Corrections within the database
- Setting up and changing user hierarchy and authority
- Review and manipulation of all data in the system

Supervisor

- ID
- Password
- Time and Date
- **Email:** idriz-konorad@bge.de
- **Name:** Idriz
- **Surnem:** Javric
- **Address:** Mühlenweg 12, 33098 Paderborn
- **Supervisor:** True

User authorizations

- Everything a employee can do, plus:
- Overview of statistical data, including the amount of free space in pre-storage and storage
- Receiving alerts about storages overfill
- Overview of special warnings such as increased levels of radioactivity, waste leakage

Employee

- ID
- Password
- Time and Date
- **Email:** kris-konorad@bge.de
- **Name:** Kris
- **Surname:** Schmidt
- **Address:** Mühlenweg 12, 33098 Paderborn
- **Employee:** True

User authorizations

- Entering data for the type of waste